Bill of Materials

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Project Title: Bill of Materials for Project [high_voltage_controller.PrjPcb] (No PCB Document Selected)
Project File Name: high_voltage_controller.PrjPcb
Assembly Variant: None

Part Number	Description	Designator	Manufacture	Manufacture Part Number	Supplier	Supplier Part Number	Supplier Unit Price	Quantity
CC0805KKX5R5BB	Surface Mount Ceramic Multilayer Capacitor 4.7uF 10% X5R 6.3V 0805 Blister T/R	C1, C20, C22			DigiKey	311-1861-1-ND		3
CC0603MRX5R5B	6.3V 1uF X5R ±20% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS, 16V 1uF X7R ±10% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C2, C10, C12, C15, C17			DigiKey	311-1813-1-ND, 311-1446-1-ND		5
CC0603KRX7R9B	Multilayer Ceramic Capacitor, CC Series, 0.1 - F, ± 10%, X7R, 50 V, 0603 [1608 Metric]	C3, C4, C5, C8, C9, C18, C19, C21, C23, C24			DigiKey	311-1344-1-ND		10
CC0603MRX7R9B	50V 10nF X7R ±20% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C6			DigiKey	311-1826-1-ND		1
CC0805KKX5R5B	6.3V 1uF X5R ±10% 0805 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C7			DigiKey	311-3417-1-ND		1
C0805C472KDRA	Cap Ceramic 0.0047uF 1000V X7R 10% Pad SMD 0805 125C Automotive AEC-Q200 Medical T/R	C11, C26			DigiKey	399-16659-1-ND		2
CC0603MRX5R7B	Multilayer Ceramic Capacitors MLCC – SMD/SMT; YAGEO; CC0603MRX5R7BB106; 10 uF; 0603; 16 V; 16 V	C13, C14, C16, C25			DigiKey	311-1822-1-ND		4
SP0504BAHTG	SP050xBA Series 8 V 30 pF Uni-Directional SMT TVS Avalanche Diode Array SOT-23-5	D1			DigiKey	F3157CT-ND		1
BLM21PG221SN1	Ferrite Beads Multi-Layer 220Ohm 25% 100MHz 2A 45mOhm DCR 0805 Paper T/R	FB1			DigiKey	490-1054-1-ND		1
FTSH-105-01-L-D	CONN HEADER SMD 10POS 1.27MM / Connector Header Surface Mount 10 position 0.050" (1.27mm)	J1			DigiKey	SAM8799-ND		1
618015231121	15 Way Right Angle Through Hole Pcb D-sub Connector Socket, 2.77MM, 2.84MM Pitch	J2			DigiKey	732-618015231121-ND		1
USB4105-GF-A	USB Connector, 24 Contact(s), Female, Right Angle, Solder Terminal, Locking, Receptacle	J3			DigiKey	2073-USB4105-GF-ACT-ND		1
RFM-0505S	DC DC CONVERTER 5V 1W / Isolated Module DC DC Converter 1 Output 5V 200mA - 5V Input	PS1			Future Electronics	RFM-0505S		1
2N7002K	Mosfet N-ch 60V 300MA SOT23 / Trans Mosfet N-ch 60V 0.3A 3-PIN SOT-23 T/r	Q1, Q2, Q3			DigiKey	2N7002KCT-ND		3
RC0805FR-0722R	125mW Thick Film Resistors 150V ±100ppm/℃ ±1% 22 Ω 0805 Chip Resistor - Surface Mount ROHS	R1			DigiKey	311-22.0CRCT-ND		1
RC0805FR-0710K	Res General Purpose Thick Film 0805 10k Ohm 1% 1/8W ±100ppm/°C Ceramic Paper T/R	R2, R7, R8, R10, R11			DigiKey	311-10.0KCRCT-ND		5
RC0805FR-07100	125Mw Thick Film Resistors 100Ppm/ 1% 100K 0805 Chip Resistor Surface Mount Rohs	R3, R6			DigiKey	311-100KCRCT-ND		2
RC0805FR-07300	Res Thick Film 0805 300K Ohm 1% 0.125W(1/8W) ±100ppm/C Pad SMD T/R	R4, R5			DigiKey	311-300KCRCT-ND		2
RC0603FR-071M	Res Thick Film 0603 1M Ohm 1% 0.1W(1/10W) ±100ppm/°C Pad SMD T/R / RES SMD 1M OHM 1% 1/10W 0603, 125mW Thick Film Resistors 150V ±100ppm/℃ ±1% 1MΩ 0805 Chip Resistor - Surface Mount ROHS	R9, R17			DigiKey	311-1.00MHRCT-ND, 311-1.00MCRCT-ND		2
RC0805FR-075K1	125Mw Thick Film Resistors 100Ppm/ 1% 5.1K 0805 Chip Resistor Surface Mount Rohs	R12, R14			DigiKey	311-5.10KCRCT-ND		2
RC0805FR-0722K	Res Thick Film 0805 22.1K Ohm 1% 0.125W(1/8W) ±100ppm/C Pad SMD T/R	R13			DigiKey	311-22.1KCRCT-ND		1
RC0805FR-0747K	47.5 kOhms ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thick Film	R15			DigiKey	311-47.5KCRCT-ND		1
RC0805FR-071KL	Resistor, thick film, SMD 0805 (2012), 1 kΩ, 0.125 W, ±1 %, RC0805FR-071KL	R16			DigiKey	311-1.00KCRCT-ND		1
STM32G431C6T6	Microcontroller 32-Bit ARM Cortex-M4 RISC 32K-Byte Flash 3.3V 48-Pin LQFP	U1			DigiKey	STM32G431C6T6-ND		1
TLV9002QDRQ1	IC OPAMP GP 2 CIRCUIT 8SOIC / General Purpose Amplifier 2 Circuit Rail-to-Rail 8-SOIC	U2			DigiKey	296-TLV9002QDRQ1CT-ND		1
TLV75533PDBVR	500mA high-PSRR low-IQ low-dropout voltage regulator with enable 5-SOT-23 -40 to 125	U3			DigiKey	296-50411-6-ND		1
ISO7721FDWR	Robust EMC, dual-channel, 1/1, reinforced digital isolator 16-SOIC -55 to 125	U4			DigiKey	296-50346-1-ND		1
CP2102N-A02-GQ	IC Usb to Uart Bridge QFN24 / Usb Bridge, Usb to Uart Usb 2.0 Uart Interface 24-QFN (4X4)	U5			DigiKey	336-5888-1-ND		1
							$0.00	57

Approved	Notes